Who benefits most?

Sorted partial effects · A colon cancer trial, R code and inference

Valentin Klotzbücher · Department of Clinical Research

Session overview

1. Literature and software.
2. A public colon cancer trial.
3. Sorted effects and characteristics of the tails.
4. Inference, validation and discussion.

Literature

01

The average effect is useful. What else can we learn from the same model?

↓ The method and its R implementation

Two starting points

- **Applied tutorial:** *The R Journal* (Chen et al. 2020).
- **Method:** *Econometrica* (Chernozhukov, Fernández-Val, and Luo 2018).

Average effect

How large is the effect on average?

Sorted effects

How much do the fitted effects vary?

Classification

Who is in the low and high tails?

What the R package provides

`spe()`

An effect distribution and average effect, with **bootstrap confidence bounds**.

`ca()`

Tail comparisons, with **bootstrap standard errors and joint p-values**.

Inference is implemented. Numerical checks still matter.

SortedEffects 1.7.0. Colon SPE uses the original package; CA inference uses a separately audited numerical stabilization. Details in backup.

The trial

02

Does chemotherapy reduce mortality by the same amount for every patient profile?

↓ A public randomized trial and a defined outcome

A public colon cancer trial

- Patients with **resected colon cancer**, randomized to three arms.
- We compare **levamisole + 5-FU** with **observation**: 619 patients.
- Outcome: **death within three years**; known for 618 patients.
- Primary analysis: **593 patients** with complete baseline variables.

Public `survival::colon` data; one death record per patient. Levamisole-only arm excluded. Trial reports: Moertel et al. (1990, 1995), PMIDs 2300087 and 7847642.

Start with the average

Observation

34.7%
109 deaths / 314 patients

Levamisole + 5-FU

25.7%
78 deaths / 304 patients

9.1 percentage points lower mortality. How much might benefit vary?

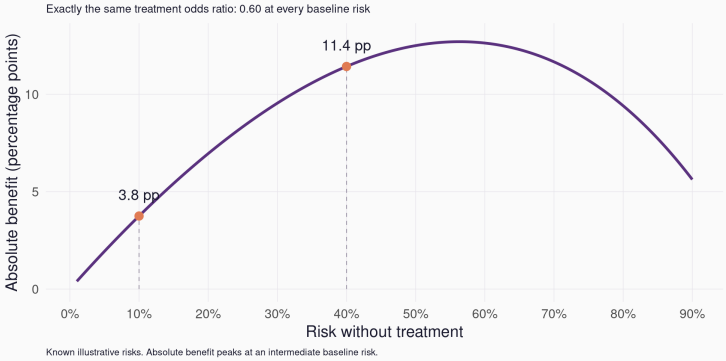
Crude three-year risks among 618 patients with known outcomes; subsequent models use 593 complete-baseline patients.

Define benefit before sorting

$$\Delta(x) = \Pr(Y = 1 \mid A = 0, X = x) - \Pr(Y = 1 \mid A = 1, X = x).$$

- $Y = 1$: death by three years; $A = 0$: observation; $A = 1$: chemotherapy.
- X: age, sex and tumour characteristics at baseline.
- **Positive:** lower mortality under treatment. **Negative:** higher mortality.
- One conditional **average** contrast per profile, not an individual's causal effect.

Effect scale matters



An algebraic illustration, not an estimate from the colon trial.

- Treatment odds ratio: **0.60** throughout.
- Control risks of **10% and 40%** give reductions of **3.8 and 11.4 pp**.
- Absolute benefit can vary without a treatment interaction on log odds.

Estimated effects

03

Fit the model, predict both conditions, then sort the risk differences.

↓ R code and the sorted-effects curve

Choose a working model

```r
source("R/colon-data.R")
source("R/colon-analysis.R")
dat <- prepare_colon_data()$data
fm <- colon_formulas()$interaction
fit <- glm(fm, data = dat, family = binomial())
```

- **Comparison model:** one treatment odds ratio, adjusted for baseline risk.
- **Interaction model:** adds treatment × age, sex, age × sex, nodes, grade and extent.
- Both specifications were written down before fitting this pilot.

Full formula in backup; executable calculations in `R/colon-walkthrough.R`.

Predict both treatment conditions

```r
control <- treated <- dat
control$treatment <- "observation"
treated$treatment <- "chemotherapy"

benefit <- predict(fit, control, type = "response") -
  predict(fit, treated, type = "response")
```

- Keep every patient's baseline characteristics fixed.
- Change only treatment assignment.
- Subtract predicted **risks**, not log odds.

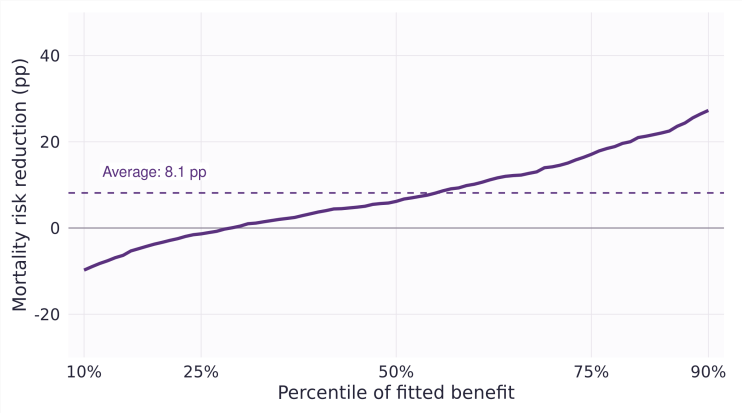
Calculate percentiles and the mean

```
u <- seq(.10, .90, by = .01)
spe_curve <- Hmisc::wtd.quantile(
  benefit, weights = rep(1, nrow(dat)), probs = u
)
mean(benefit)
```

- **SPE curve:** percentiles of fitted risk reductions.
- **Average effect:** mean across all 593 profiles.
- Sorting keeps the effect values; covariate labels disappear from the axes.

Equal profile weights; the quantile convention matches SortedEffects 1.7.0.

The sorted-effects curve



- Mean benefit: **8.1 pp**.
- 10th percentile: **−9.8 pp**.
- 90th percentile: **27.3 pp**.
- A wide fitted spread. We still need uncertainty and validation.

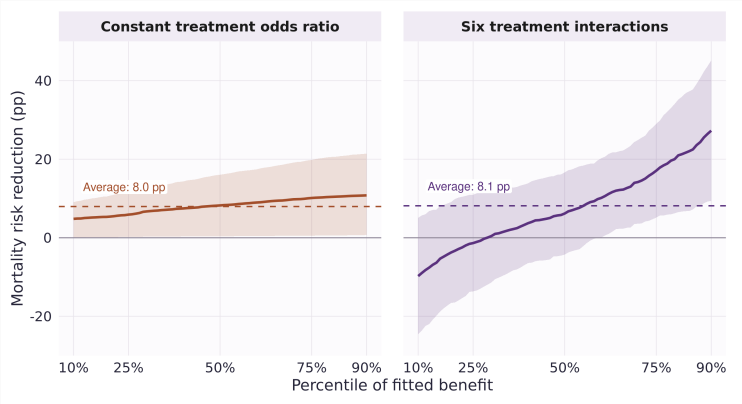
How to read one point

At the **80th percentile**, estimated benefit is **20.0 pp**.

- About **80% of profiles** have fitted benefits at or below this value.
- About **20%** have larger fitted benefits.
- A percentile **of effects**, not an effect on an outcome percentile.

It does not mean that a patient has an 80% chance of benefiting.

Similar averages, different distributions



- Mean benefit: **8.0 vs 8.1 pp**.
- Interactions widen the fitted distribution.
- Six-term interaction test: **p = .014**.
- More spread alone does not validate a ranking.

Native 95% uniform bands, calibrated separately for each model over the 10th–90th percentiles.

Classification

04

Which patient profiles sit in the two tails?

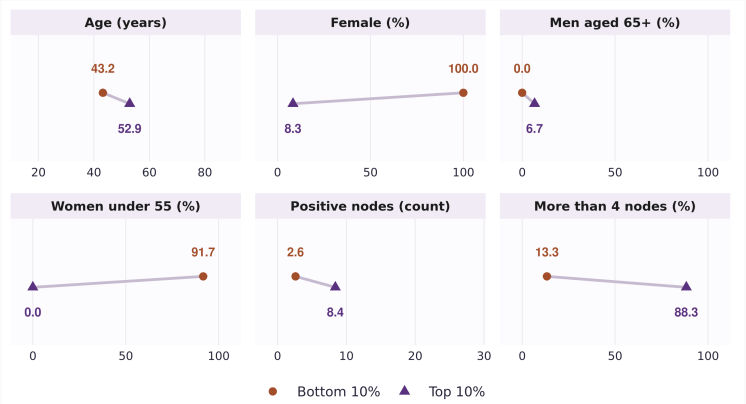
↓ Twelve characteristics, in their original units

Describe the two tails

```
cutoff <- Hmisc::wtd.quantile(
  benefit, rep(1, nrow(dat)), c(.10, .90)
)
tails <- list(low = benefit <= cutoff[1],
              high = benefit >= cutoff[2])
traits <- colon_characteristics()$variable
lapply(tails, function(i)
  colMeans(dat[i, traits]))
```

- **60 patients per tail**, selected by estimated benefit.
- Each tail can contain patients from either randomized arm.
- The model learned these groups using outcomes from this sample.

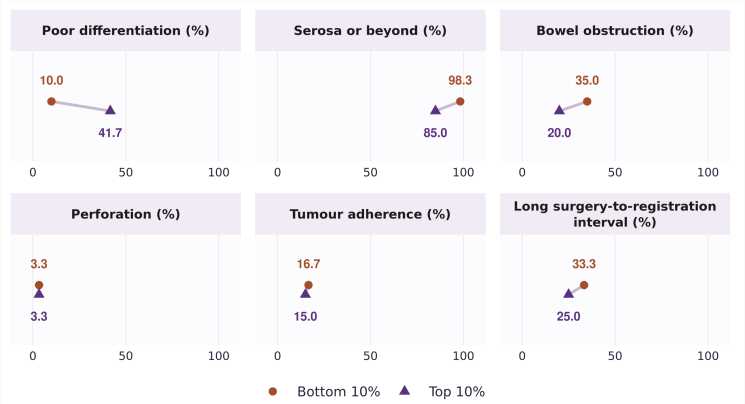
Age, sex and positive nodes



- Low tail: **all women**, mean age **43**.
- High tail: **92% men**, mean age **53**.
- More than four nodes: **13% vs 88%**.
- The high tail is not mainly older men.

Descriptive means; 60 patients per tail. Lines join two means, not confidence limits.

Other tumour and clinical characteristics



- Poor differentiation: **10% vs 42%**.
- Bowel obstruction: **35% vs 20%**.
- Perforation: **3% in both tails**.
- Visible differences still need uncertainty estimates.

Descriptive means. “Adherence” means tumour adherence to nearby organs, not adherence to treatment.

What does “mostly younger women” tell us?

- Patients were ranked using **age, sex and tumour characteristics together**.
- Younger women make up most of the fitted bottom tenth.
- This describes a **model-selected group**, not the treatment effect in all women.
- Assessing harm requires uncertainty for a defined treatment contrast.

Classification answers “who is in this tail?” It does not test “does treatment harm women?”

Inference and validation

05

How much of the pattern could be sampling variation or model choice?

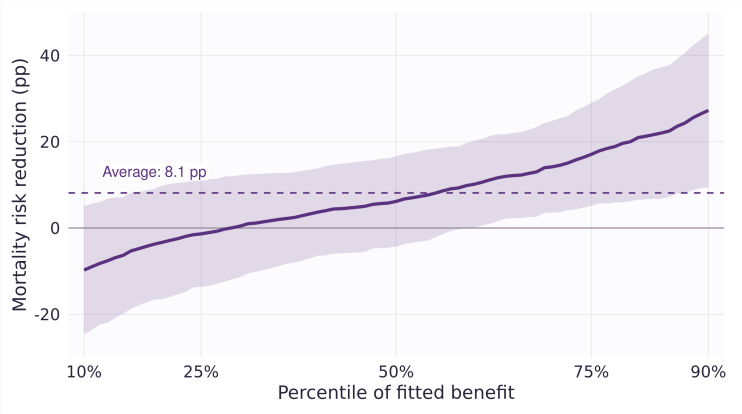
↓ Bands, classification inference and a held-out check

Ask the package for a confidence band

```r
spe_result <- SortedEffects::spe(
  fm = fm, data = dat, method = "logit",
  var_type = "categorical", var = "treatment",
  compare = c("chemotherapy", "observation"),
  us = seq(.10, .90, .01), b = 499,
  alpha = .05, bc = FALSE, seed = 420
)
```

- Refit with bootstrap weights and recompute the effect curve.
- A **uniform band** covers the displayed grid jointly, under the method's assumptions.
- This accounts for sampling variation, not every source of model error.

Negative estimates do not establish harm



- 10th percentile: **−9.8 pp**.
- Band at that point: **−24.6 to +5.1 pp**.
- The sign is uncertain at the lower end.
- Mean benefit: **8.1 pp**, CI **1.4 to 14.9**.

Native 95% uniform SPE band over the 10th–90th percentiles; separate 95% interval for the average.

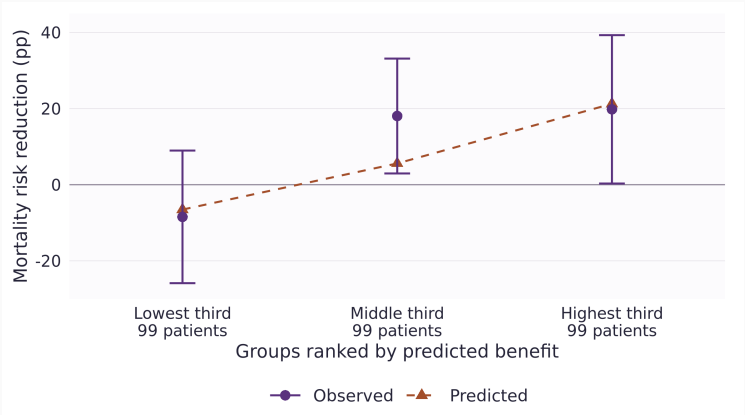
Are the tail characteristics different?

Characteristic	Low tail	High tail	Joint p
Female	100%	8%	.042
Women under 55	92%	0%	.040
More than four nodes	13%	88%	.132
Poor differentiation	10%	42%	.948

- Joint comparisons account for **all twelve specified characteristics**.
- The two small p-values concern closely related traits.

Audited stabilized CA reconstruction, 499 draws. Native CA was numerically unstable. The full table is in backup.

Does the ranking work in held-out patients?



- Fit on **296**; freeze predictions for **297**.
- One split; three groups of **99**.
- Ranking slope: **1.10**, CI **0.24–1.97**; adjusted **p = .025**.
- Low-third benefit: **−8.5 pp**, CI **−25.9 to +9.0**.

Pointwise, unadjusted 95% intervals in the plot. Slope p-value: Holm adjustment over the two planned rankings.

A promising signal, with important limits

- **Missing-data sensitivity:** a simple fill-and-indicator analysis of all 618 known outcomes changes interaction p from **.014 to .054**.
- **One held-out split:** useful, but not external validation.
- **Small tails and flexible modeling:** 60 patients per tail, 19 parameters.
- **Exploratory search:** our third trial pilot; no adjustment across datasets.

The ranking deserves follow-up. These results do not support a treatment rule or a claim of harm in women.

A useful comparison: the stroke trial

Check	Aspirin in IST	Colon chemotherapy
Estimated average benefit	1.3 pp	8.1 pp
Joint treatment-interaction p	.355	.014
Held-out interaction-ranking p	.822	.013

- IST also produced variation in fitted effects.
- Its interaction-model ranking did **not** carry over to held-out patients.
- A broad curve is a reason to investigate, not a validation result.

Different endpoints: IST six-month death/dependence; colon three-year mortality. Raw within-pilot p-values shown; colon ranking Holm p = .025.

Discussion

06

What would make a sorted-effects analysis useful in our own work?

↓ Interpretation, credibility and next steps

What did sorting add?

- Two models gave almost the **same average benefit**, but different effect distributions.
- Classification made the tails concrete: age, sex and tumour features together.
- Uncertainty and held-out results changed how strongly we could describe the pattern.

Which result would you want to see before taking this pattern seriously?

Where do causal forests fit?

Causal forests	Sorted effects + classification
Estimate how treatment effects vary across patient profiles. Many trees learn useful splits and interactions in the baseline characteristics.	**Summarize** the distribution of fitted effects and who lies in its tails. Sort the estimates, then compare low- and high-effect profiles.

- A forest could replace our regression model; sorting asks the same descriptive questions.
- Forest estimates need suitable inference for **bands and tail comparisons**.

Generalized random forests (Athey, Tibshirani, and Wager 2019). A conceptual connection; no forest was fitted for this example.

Reading and reproducible materials

- **Send ahead:** the R tutorial, introduction and applied example (Chen et al. 2020).
- **Optional theory:** the original method paper (Chernozhukov, Fernández-Val, and Luo 2018).
- **Our example:** colon report with the full table, validation, sensitivity checks and data provenance.
- **Code and slides:** github.com/valentinklotzbuecher/jcod.

Additional material

+

Full tables, model specification and software checks.

↓ Optional slides for questions

Backup · What is being sorted?

$$\mathsf{SPE}(u) = Q_{\Delta(X)}(u), \qquad \mathsf{APE} = E[\Delta(X)].$$

- The distribution of X defines the population summarized.
- Classification describes covariates in selected tails of $\Delta(X)$.
- A quantile of effects differs from a difference of outcome quantiles.
- Randomization supports identification; sorting does not replace it.

Backup · The full interaction model

```
outcome ~ treatment + splines::ns(age, df = 3) + female +
  age_z:female + log_nodes + poor_diff + extensive +
  obstruct + adhere + surg_late +
  I((treatment == "chemotherapy") * age_z) +
  I((treatment == "chemotherapy") * female) +
  I((treatment == "chemotherapy") * age_z * female) +
  I((treatment == "chemotherapy") * log_nodes) +
  I((treatment == "chemotherapy") * poor_diff) +
  I((treatment == "chemotherapy") * extensive)
```

`age_z = (age - 60) / 10`; `log_nodes = log1p(nodes)`.

Binomial logistic regression; 19 parameters. Comparison model omits the final six terms.

Backup · Classification: demographics and nodes

Characteristic	Low	High	Difference	SE	Joint p
Age (years)	43.2	52.9	+9.7	7.1	0.872
Female (%)	100.0	8.3	-91.7	21.4	0.042
Men aged 65+ (%)	0.0	6.7	+6.7	14.2	1.000
Women under 55 (%)	91.7	0.0	-91.7	21.1	0.040
Positive nodes	2.6	8.4	+5.8	2.2	0.297
More than 4 nodes (%)	13.3	88.3	+75.0	22.1	0.132

Difference = high − low. Percentages differ in percentage points.

60 patients per tail. SEs and joint p-values: audited stabilized CA, 499 draws. Adjustment covers all twelve traits, not just these six.

Backup · Classification: clinical characteristics

Characteristic	Low	High	Difference	SE	Joint p
Poor differentiation (%)	10.0	41.7	+31.7	27.7	0.948
Serosa or beyond (%)	98.3	85.0	-13.3	17.4	0.992
Bowel obstruction (%)	35.0	20.0	-15.0	9.6	0.762
Perforation (%)	3.3	3.3	+0.0	3.7	1.000
Tumour adherence (%)	16.7	15.0	-1.7	10.5	1.000
Long surgery interval (%)	33.3	25.0	-8.3	10.8	0.990

Difference = high − low, in percentage points.

“Long surgery interval”: surgery-to-registration category. SEs and joint p-values: audited stabilized CA, adjusted across all twelve traits.

Backup · Calling classification analysis

```
ca_result <- SortedEffects::ca(
  fm = fm, data = dat, method = "logit",
  var_type = "categorical", var = "treatment",
  compare = c("chemotherapy", "observation"), u = .10,
  interest = "moment", cl = "diff",
  t = colon_characteristics()$variable,
  b = 499, alpha = .05, bc = FALSE, seed = 420
)
```

This native call is numerically unstable for the colon model.

Raw output preserved separately. Endorsed CA inference is reconstructed by `scripts/verify-colon-inference.R stabilize` after the native audit.

Backup · What went wrong in native CA?

- CA rescales bootstrap fitting weights to total **20,000**.
- **141/499 fits did not converge**; 137 draws had undefined tail moments.
- The routine still returned finite-looking inference, which we do not use.
- Stable starting coefficients recovered **499 converged, finite draws**, with none discarded.

Local reconstruction preserves resamples, model, threshold definitions and moment formulas. The original installed package was unchanged.

Backup · What is resampled?

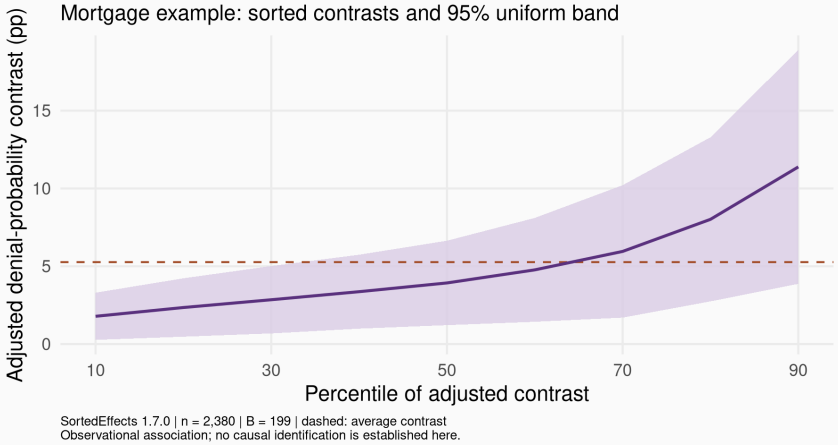
- `spe()` **full sample:** refits with bootstrap weights; effect quantiles retain original profile weights.
- `ca()` **moments:** refits and updates thresholds and characteristic means with resampled weights.
- **Held-out validation:** fits on training data; evaluates the frozen ranking using test outcomes.

Backup · Cohort and sensitivity checks

Analysis	Patients	Average benefit	Interaction p
Complete baseline and endpoint	593	8.1 pp	.014
Missing baseline values with indicators	618	8.4 pp	.054
Early-censored outcome assigned 0	594	8.0 pp	.017
Early-censored outcome assigned 1	594	8.4 pp	.010

- The latter two retain complete baseline variables.
- Eight records disagree on the supplied vs count-derived four-node indicator; the report includes a coding check.

Backup · Original published example



Tutorial mortgage data (Munnell et al. 1996; Chen et al. 2020). Separate original-package benchmark; adjusted observational disparity.

References

Athey, Susan, Julie Tibshirani, and Stefan Wager, “Generalized random forests,” *The Annals of Statistics*, 47 (2019), 1148–1178.

Chen, Shuowen, Victor Chernozhukov, Iván Fernández-Val, and Ye Luo, “SortedEffects: Sorted causal effects in R,” *The R Journal*, 12 (2020), 131–146.

Chernozhukov, Victor, Iván Fernández-Val, and Ye Luo, “The sorted effects method: Discovering heterogeneous effects beyond their averages,” *Econometrica*, 86 (2018), 1911–1938.

Munnell, Alicia H., Geoffrey M. B. Tootell, Lynn E. Browne, and James McEneaney, “Mortgage lending in Boston: Interpreting HMDA data,” *The American Economic Review*, 86 (1996), 25–53.